FINANCIAL RISK ANALYSIS REPORT

PREDICTING NEXT-QUARTER DEBT SERVICING RISK FOR PHILIPPINE COMPANIES

CORPORATE SOLVENCY AND PREDICTIVE MODELING

A Presentation by Jean Shermin B. Geronimo

Predicting Next-Quarter Debt Servicing Risk for Philippine Companies

CORPORATE DEBT SERVICING CHALLENGES

THE BUSINESS PROBLEM

- Listed non-financial firms face critical debt servicing risks requiring timely identification.
- Traditional ratios assume linear relationships, failing to capture complex risk patterns.
- Manual review processes are too time-consuming for proactive risk mitigation.

NECESSITY OF AI-DRIVEN SYSTEMS

- AI detects subtle, non-linear interactions among diverse financial indicators.
- Provides an early warning system to overcome limitations of manual oversight.

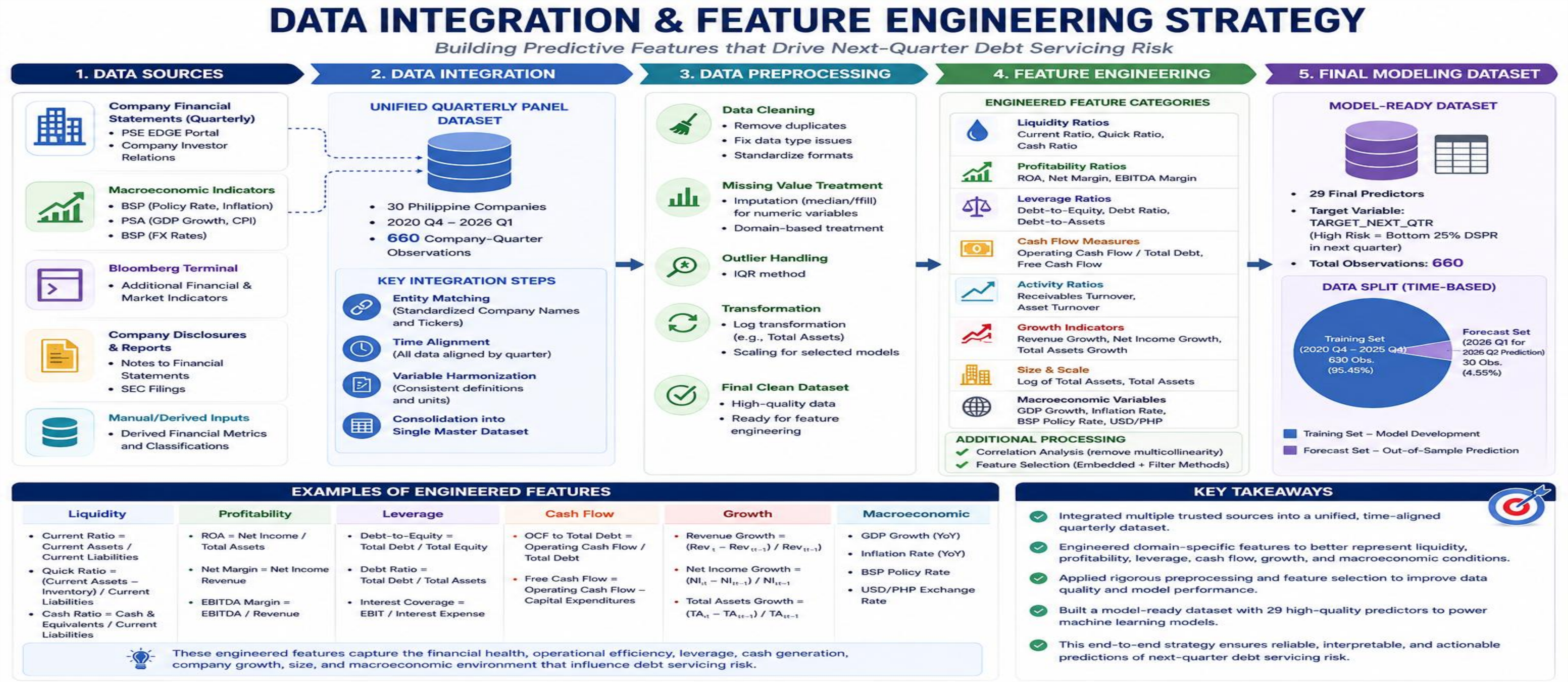
MACROECONOMIC CONTEXT

- Rising interest rates and inflation pressure corporate repayment capacities.
- Exchange rate fluctuations (USD/PHP) significantly impact debt servicing costs.

STUDY PARAMETERS

Bottom 25%
DSPR Threshold for "High Risk" Classification

2020 Q4 – 2026 Q1
Quarterly Time Horizon for Predictive Analysis

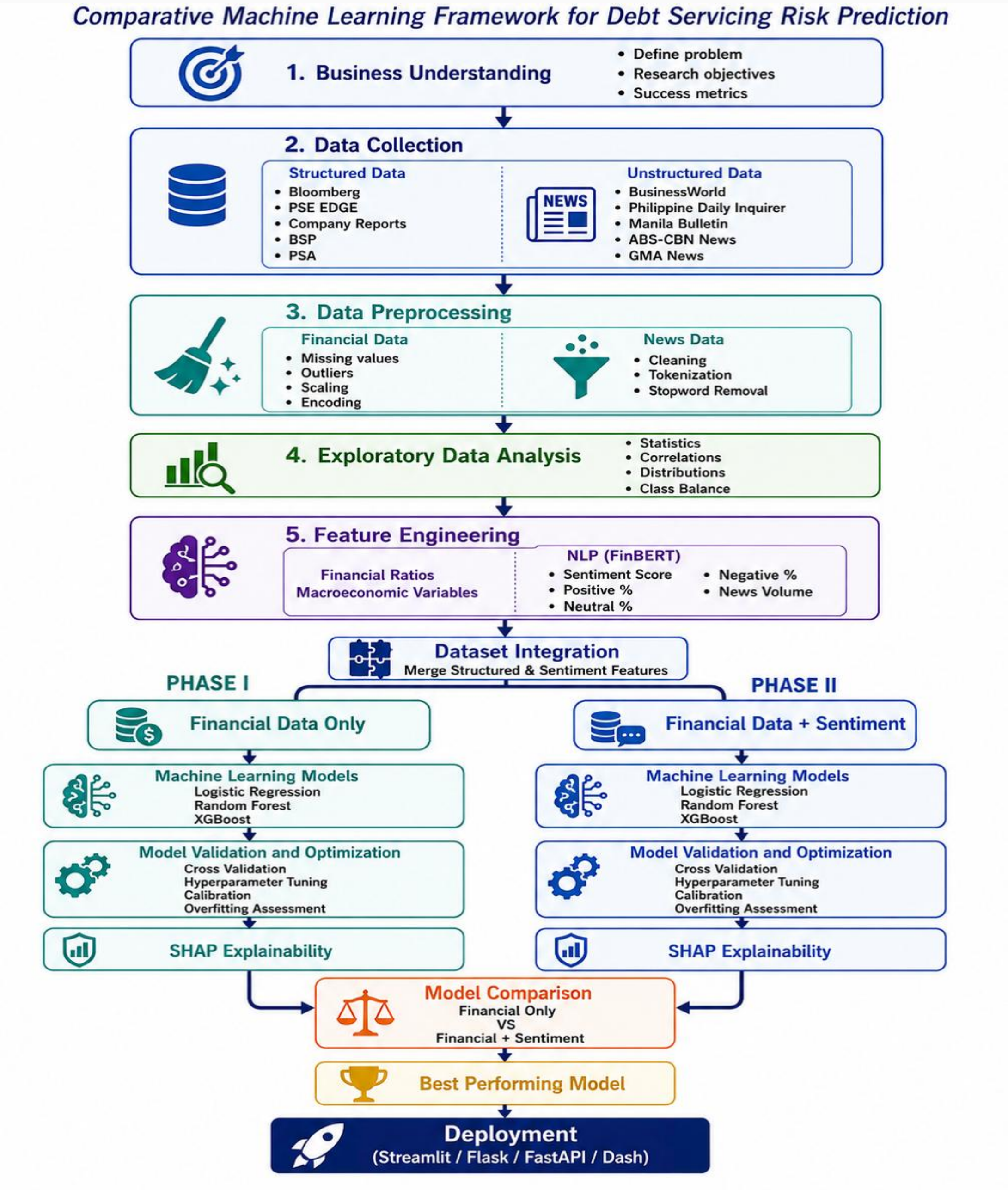


STRUCTURED PANEL DATASET: Integrated financial data from Bloomberg Terminal and official Philippine sources including PSE EDGE, PSA, and BSP.

DATASET SCOPE: Comprises 660 company-quarter observations for 30 firms spanning 2020 Q4 to 2026 Q1, totaling 38 curated variables.

ENGINEERED INDICATORS: Key metrics include liquidity, profitability, and leverage ratios, paired with macro indicators like GDP Growth, Inflation, and USD/PHP rates.

SHIFT LOGIC & LEAKAGE PREVENTION: Financial data from quarter *t* predicts risk for *t+1*, strictly excluding future-revealing variables (DSR_INT, TARGET_NEW, STDO).



MACHINE LEARNING FRAMEWORK

Supervised binary classification framework to predict next-quarter (TARGET_NEXT_QTR) debt servicing risk based on historical financial indicators.

Model Selection

Logistic Regression: Interpretable baseline with L1/L2 regularization.

Random Forest: Robust ensemble against noise.

Advanced Boosting

XGBoost: High-performance gradient boosting with built-in regularization and superior scalability for financial time-series.

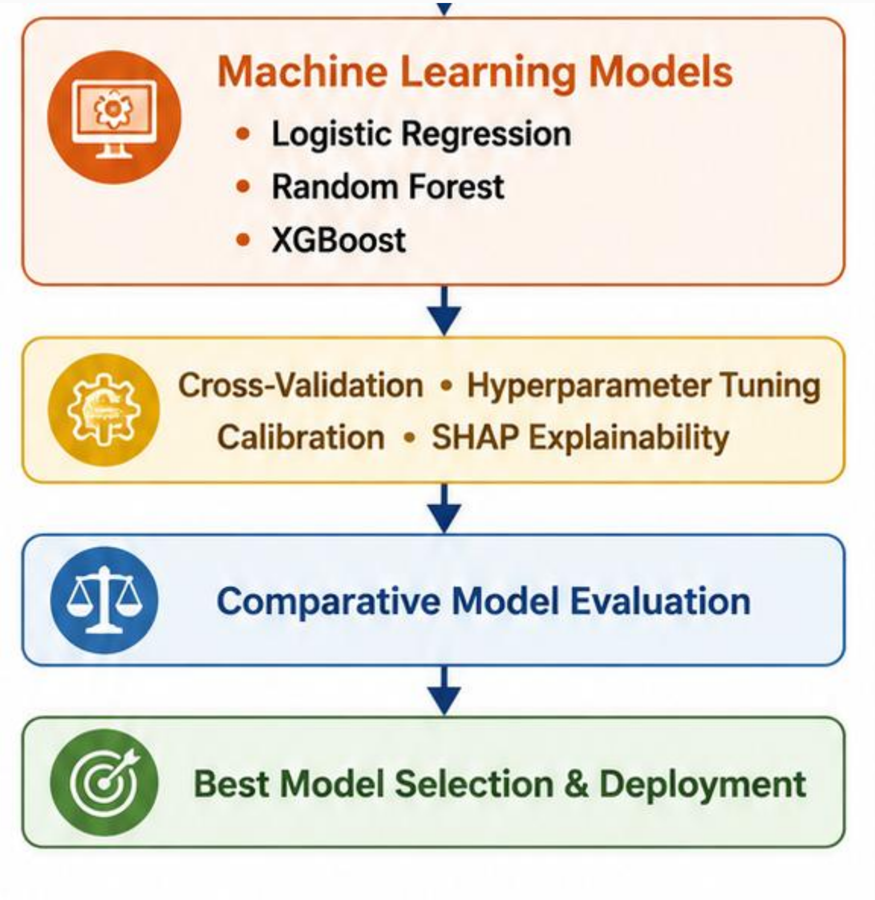
Data Strategy

Chronological Split: Training 2020 Q4–2025 Q4. Uses **TimeSeriesSplit** to prevent look-ahead bias in forecasting.

Success Metrics

Primary: ROC-AUC for discriminant ability. Supported by Brier Score, F1 Score, and Generalization Gap analysis.

PREDICTIVE MODELING & TUNING



Model Implementation:

- **Logistic Regression:** Robust baseline using L1/L2 regularization for interpretability.
- **Ensemble Methods:** Random Forest (Bagging) and XGBoost (Gradient Boosting) for non-linear pattern detection.

Preprocessing & Tuning:

- **Data Cleaning:** Median imputation for missing values and StandardScaler for feature scaling.
- **Optimization:** RandomizedSearchCV utilized for efficient hyperparameter cross-validation ROC-AUC tuning.

Performance Optimization:

- **Overfitting Control:** Implementation of Alpha and Lambda regularization in boosting models.
- **Strategic Trade-off:** Balancing baseline transparency with high-accuracy predictive robustness.

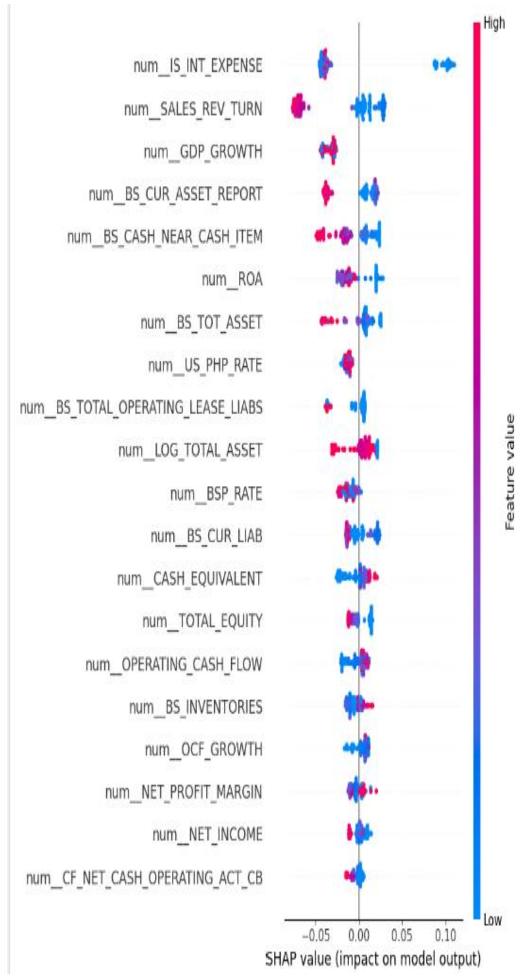
MODEL INTERPRETATION VIA EXPLAINABLE AI (XAI)

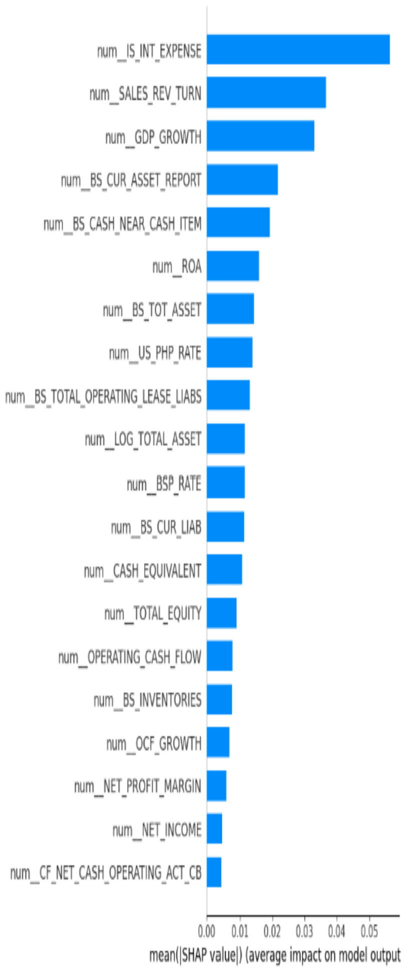
Transparency: XAI provides observation-level transparency via SHAP values, detailing feature contributions to "black-box" model predictions.

Methodology: Contrasts with global methods like MDI or Gain-based importance which offer general significance but lack individual case precision.

Business Value: Builds stakeholder trust, enhances credit evaluation, and enables proactive risk management while ensuring accountability.

Compliance: Aligns with OECD AI Principles and NIST AI Risk Management Framework regarding automated decision transparency.





STRATEGIC IMPLEMENTATION & STAKEHOLDER IMPACT

EMPOWERING STAKEHOLDERS WITH PROACTIVE INTELLIGENCE

- **Financial Institutions:** Enhance credit risk assessment and loan monitoring for high-risk borrowers.
- **Investors:** Earlier detection of vulnerability to improve portfolio risk and decision-making.
- **Corporate Managers:** Strengthen health monitoring and optimize financial planning/corrective measures.

DEPLOYMENT-READY REAL-TIME MONITORING

- Deployable ML model integrated via Flask or Streamlit for continuous real-time risk monitoring.
- Leverages GDP Growth, BSP Rates, Inflation, and USD/PHP Exchange for comprehensive insights.

DRIVING EFFICIENCY & INFORMED DECISIONS

- Significant reduction in manual effort for financial analysis and credit risk assessment.
- Prioritized monitoring of high-risk entities with capability to predict 2026 Q2 debt servicing risk.